

Industry Overview:

With the implementation of AI technologies into the healthcare industry, new functionalities emerge in several parts of hospitals around the world. Virtual doctor/nurse visits have become more common, the detection of diseases by AI analysis is becoming more widespread, and the management of hospital data by an AI system is already active in many large hospitals. Combined with simple blockchaining techniques, hospitals have achieved efficiency that was thought to be impossible in the decades before automation.

Current Trends:

Blockchaining in the medical field can include many things, like pharmaceutical supply chains, remote patient monitoring, or even electronic health records. For pharmaceutical companies, blockchain techniques have been implemented to combat the severe drug counterfeit and regulation issues. In this case, using blockchaining, these companies were able to develop a drug standardization process which analyzes each

prescribed drug and determines the validity of them. Counterfeit or defective drugs would be able to be identified and removed before distribution. For remote patient monitoring, blockchaining is responsible for the storage, upkeep, and extraction/retrieval of patient data gathered from monitoring.

Enhancing Clinical and Administrative Efficiency:

Beyond data management, blockchain can optimize several operational processes:

- Smart contracts automate insurance claims, billing, and authorization workflows.
- Decentralized identity verification reduces fraud in patient registration and provider credentialing.
- Supply chain tracking ensures authenticity of pharmaceuticals and medical devices.

These improvements reduce costs, minimize human error, and increase transparency across the healthcare ecosystem.

Supporting AI-Driven Healthcare Systems:

Earlier sections describe the rise of AI in diagnostics, virtual care, and hospital operations. Blockchain complements these systems by:

- Providing secure datasets for AI training.
- Ensuring that AI-generated outputs (diagnoses, risk scores, treatment plans) are logged immutably.
- Preventing data poisoning attacks that could compromise AI reliability.

The synergy between AI and blockchain creates a more trustworthy environment for automated decision-making—critical in high-stakes medical contexts.

Challenges and Limitations:

Despite its promise, blockchain adoption in healthcare faces several obstacles:

- Scalability issues when handling large medical datasets (e.g., imaging).

- High computational costs for maintaining distributed ledgers.
- Regulatory uncertainty around decentralized data storage.
- Integration complexity with legacy hospital systems.

The article notes that hybrid models—where blockchain stores metadata and access logs while traditional databases store large files—are emerging as a practical solution.

Conclusion:

Blockchain technology is steadily transforming healthcare by reinforcing security, enhancing interoperability, and empowering patients with greater control over their medical data. When combined with AI-driven diagnostics, remote monitoring systems, and automated clinical workflows, blockchain provides the structural backbone needed to support safe, efficient, and transparent digital healthcare ecosystems. Although challenges remain—particularly around scalability and regulatory alignment—the trajectory of current research and industry

adoption suggests that blockchain will play a central role in the next generation of healthcare infrastructure.

Sources:

https://pmc.ncbi.nlm.nih.gov/articles/PMC10701638/#sec1_8